

CMSE 801 - Day 2



Order of magnitude modeling using Python

August 28, 2024

General Course Announcements

- There is **no class this coming Monday, September 2**. It is a University holiday for Labor Day.
- The Day 3 Pre-class Assignment might feel like **a lot**, but you have more time to complete it than a normal pre-class assignment. This is an opportunity to start pushing yourself with your new Python skills -- it's OK to get stuck!
 - In general, PCAs should take you somewhere between 30 and 90 minutes, but there will be some variation from one to the next. You can also bring questions you have about the PCAs to help room/office hours, as needed, but remember that the PCAs are not graded for correctness.
- Remember: at the end of class **you will be uploading your in-class notebook to D2L**.

Questions/comments from Day 2 PCA

- Some general challenges with getting used to using Jupyter/Python and being new to this type of work.
- When should I create more cells in my notebook? Are there limits on how much code I can put in one cell?
- What is the deal with strings and why are they useful?
 - Why would I ever want to turn a number into a string?
- A number of questions came up related to Section 2.3 and Question 3 -- **we'll dig into some of that right now!**
- Also some comments/questions about Question 4 and how to solve this sort of problem -- **You'll be practicing these sorts of things in class today!**

```
In [1]: # Define some variables with numbers in them, then print the result of s  
x = 4.2  
y = 4.3  
print(x+y)  
  
# I could also store this result in a new variable  
z = x+y  
print(z)
```

8.5

8.5

```
In [2]: # What if I didn't use the "print" command in the previous example?  
x = 4.2  
y = 4.3  
x+y  
  
z = x+y  
z
```

```
Out[2]: 8.5
```

```
In [3]: # But what if I convert these numbers to strings and then add them?  
x = str(x)  
y = str(y)  
  
z = x+y  
print(z)
```

4.24.3

That's not particularly useful for this simple example, but could I use this to create something more useful?

```
In [4]: print("The value of x is "+x+" and the value of y is "+y)
```

The value of x is 4.2 and the value of y is 4.3

But couldn't I have done that without converting them?

```
In [5]: x = 4.2
y = 4.3
print("The value of x is "+x+" and the value of y is "+y)
```

```
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TypeError                                Traceback (most recent
call last)
Cell In[5], line 3
      1 x = 4.2
      2 y = 4.3
----> 3 print("The value of x is "+x+" and the value of y is "+
y)

TypeError: can only concatenate str (not "float") to str
```

And remember, a string is just a collection/list of individual characters, so if I need to grab just one character, I can do that. (you'll learn more about lists in the next pre-class assignment)

```
In [16]: name = "Devin"  
         print(name[0])
```

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Goals for today

- Start getting comfortable writing and execute code -- **remember**: writing bugs and getting errors is part of the process!
- Using Python, the internet, and discussions with your group to **do some "order of magnitude" estimates** (sometimes called "Fermi Problems")

Let's get started

1. Download the notebook from the website
2. Work on it with your group.
 - A. We'll count off groups to determine which questions you're going to work on first.
3. **Be prepared to share out your results around 9:40am.** We likely won't have time for every group to share, so we'll just hear from 2-3 groups.

